Name: V.Mahalakshmi
Reg No: 192521181

Water Jug Problem

Two jugs of 4 gallons and 3 gallons are given, with no measuring marks. A pump with unlimited water is available to fill the jugs, and water can be poured on the ground. The aim is to get exactly 2 gallons of water in the 4-gallon jug.

1. Problem Formulation

State: (x, y), where x is the quantity of water in the 4-gallon jug (x = 0, 1, 2, 3, 4) and y is the quantity of water in the 3-gallon jug (y = 0, 1, 2, 3). There are $5 \times 4 = 20$ possible states.

Initial state: (0, 0) – both jugs are empty.

Goal state: (2, n) – exactly 2 gallons in the 4-gallon jug, with any amount n in the 3-gallon jug.

Path cost: Each rule applied costs 1, so the cost is the number of steps.

2. Production Rules (Operators)

Rule 1: (x, y) if $x < 4$ gives (4, y) – Fill the 4-gallon jug.

Rule 2: (x, y) if $y < 3$ gives (x, 3) – Fill the 3-gallon jug.

Rule 3: (x, y) if $x > 0$ gives (0, y) – Empty the 4-gallon jug on the ground.

Rule 4: (x, y) if $y > 0$ gives (x, 0) – Empty the 3-gallon jug on the ground.

Rule 5: (x, y) if $x + y \geq 4$ and $y > 0$ gives $(4, y – (4 – x))$ – Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full.

Rule 6: (x, y) if $x + y \geq 3$ and $x > 0$ gives $(x – (3 – y), 3)$ – Pour water from the 4-gallon jug into the 3-gallon jug until the 3-gallon jug is full.

Rule 7: (x, y) if $x + y \leq 4$ and $y > 0$ gives $(x + y, 0)$ – Pour all the water from the 3-gallon jug into the 4-gallon jug.

Rule 8: (x, y) if $x + y \leq 3$ and $x > 0$ gives $(0, x + y)$ – Pour all the water from the 4-gallon jug into the 3-gallon jug.

3. Solution (Breadth-First Search)

Starting from (0, 0), breadth-first search applies the rules level by level until a state with 2 gallons in the 4-gallon jug appears. The shortest path has 6 steps:

Start: (0, 0) – both jugs empty.

Step 1: apply Rule 1 – Fill the 4-gallon jug. State: (4, 0)

Step 2: apply Rule 6 – Pour water from the 4-gallon jug into the 3-gallon jug until the 3-gallon jug is full. State: (1, 3)

Step 3: apply Rule 4 – Empty the 3-gallon jug on the ground. State: (1, 0)

Step 4: apply Rule 8 – Pour all the water from the 4-gallon jug into the 3-gallon jug. State: (0, 1)

Step 5: apply Rule 1 – Fill the 4-gallon jug. State: (4, 1)

Step 6: apply Rule 6 – Pour water from the 4-gallon jug into the 3-gallon jug until the 3-gallon jug is full. State: (2, 3)

Path: (0,0) – (4,0) – (1,3) – (1,0) – (0,1) – (4,1) – (2,3)

4. Result

The 4-gallon jug holds exactly 2 gallons after 6 steps, and the final state is (2, 3). Because breadth-first search checks shorter paths first, no solution with fewer steps exists.